

pypomp: Inference for partially observed Markov process models in Python with JAX

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Abstract

Methods for fitting and evaluating partially observed Markov process (POMP) mechanistic models are useful in a variety of fields such as epidemiology, ecology, and finance, but are often hindered by high computational demands. We introduce **pypomp**, a high-performance Python package for statistical inference using POMP models. Designed for complex stochastic dynamic systems, **pypomp** implements plug-and-play, likelihood-based algorithms including sequential Monte Carlo and iterated filtering. By leveraging GPU acceleration, **pypomp** significantly outperforms its predecessor, the R package **pomp**. Models that previously required months to fit now complete in days, enabling the use of formerly impractical models. Furthermore, **pypomp** uses JAX to perform gradient descent on differentiable particle filters, achieving likelihood maximization beyond the practical limits of iterated filtering alone. The package provides a comprehensive interface that streamlines model construction, fitting, and analysis, allowing practitioners to develop and evaluate complex models with minimal implementation overhead.

Keywords: pomp, JAX, GPU acceleration, likelihood-based inference

1 Introduction

Partially Observable Markov Process (POMP) models, also known as state-space models or hidden Markov models, provide a flexible mechanistic framework for modeling time-series dynamic systems where the underlying state process is not directly observed. Methods such as particle filtering (Arulampalam et al., 2002) and iterated filtering (Ionides et al.,

2006, 2015) have been developed to perform maximum likelihood estimation and evaluation for POMP models. A defining strength of these methods is that they are performed by simulating from the process model instead of evaluating exact transition probabilities (a property known as plug-and-play (Bretó et al., 2009)), which makes them suitable for examining a wide range of models unconstrained by needs for mathematical convenience. Consequently, POMP models find extensive application in epidemiology (Mietchen et al., 2024; Fox et al., 2022; Wen et al., 2024), ecology (Auger-Méthé et al., 2021; Marino et al., 2019; Blackwood et al., 2013), finance (Bretó, 2014), and other domains where mechanistic modeling is desired.

Thus far, the **pomp** package ecosystem in R has provided a basis for POMP modeling of several types. The **pomp** package is designed for standard POMP models, while its extension packages **panelPomp**, **spatpomp**, and **phyloPomp** further enhance its capabilities for panel, spatio-temporal, and phylodynamic data analysis, respectively (King et al., 2016; Bretó et al., 2025; Asfaw et al., 2024).

While powerful, statistical inference for POMP models poses substantial computational challenges, and the existing tools are held back by their dependence on CPUs. To address the need for plug-and-play POMP methods that leverage GPU acceleration, we introduce the package **pypomp** (Abkemeier et al., 2024). Briefly stated, it is a Python implementation of the core algorithms in **pomp** and **panelPomp**, but written such that they can be run on both GPUs and CPUs. We accomplish this by using JAX (Bradbury et al., 2018), a high-performance numerical computing library that supports hardware acceleration and automatic vectorization. [AA: Need to change "90 times faster" to something more specific after we more rigorously compare pypomp with pomp. In addition, I think we should include the @korevaar2020 data set fit in this paper, as it provides perhaps the strongest motivation for developing pypomp.] The resulting package runs methods up to 90 times faster on GPUs than **pomp** can on CPUs. Jobs that would take months in **pomp** can be completed in days in **pypomp**, enabling plug-and-play likelihood-based inference on data sets that would otherwise be intractably large. This is demonstrated in Section [AA: add this section], where we fit a panel POMP model to the Korevaar et al. (2020) England and Wales measles dataset, which contains over one million observations. While this would normally take over 5 months in **pomp**, **pypomp** can finish the job in 41 hours [AA: revise these numbers as appropriate].

The remainder of this paper is organized as follows. Section 2 demonstrates the mathematical notation for POMP models and their related implementation in **pypomp**. Section 3 introduces the embedded methodologies. ?@sec-analysis presents data analysis workflows and benchmarking results. Section 4 concludes with a discussion of future work.

2 POMP Models in pypomp

A partially observed Markov process (POMP) model is defined by an initial density $f_{X_0}(x_0; \theta)$, a transition density $f_{X_n|X_{n-1}}(x_n | x_{n-1}; \theta)$, and a measurement density $f_{Y_n|X_n}(y_n | x_n; \theta)$. In **pypomp**, these are implemented as an object of class **Pomp**, which encapsulates the model mechanics and parameters.

While users can define custom models by providing JAX-compatible Python functions for each component, **pypomp** we demonstrate the API with the premade S&P500 model.

```
import pypomp as pp
import jax

spx_obj = pp.models.spx()

key = jax.random.key(1)
key, subkey = jax.random.split(key)
```

3 POMP Inference Methods

`pypomp` implements several core likelihood-based inference methods, all of which leverage the **plug-and-play property** (Ionides et al., 2006) and JAX-powered acceleration. These methods are implemented as methods of the `Pomp` class and modify the object in place by appending results to `results_history`.

1. **Particle Filter (`pfilter`)**: A standard sequential Monte Carlo (SMC) algorithm for likelihood evaluation and state estimation. It provides unbiased Monte Carlo estimates of the log-likelihood.
2. **Iterated Filtering (`mif`)**: An implementation of the IF2 algorithm (Ionides et al., 2015) for maximum likelihood estimation. It combines particle filtering with sequential random-walk perturbations and a cooling schedule to explore the parameter space.
3. **Automatic Differentiation (`mop` and `train`)**: `pypomp` introduces a differentiable variant of the particle filter (MOP- α , Tan et al. (2024)) that enables the use of JAX's automatic differentiation. The `train()` method uses these gradients (and optionally Hessians) for local optimization.

By utilizing JAX, these methods achieve high performance via JIT compilation and vectorization, allowing them to run efficiently on both CPUs and GPUs. The following example demonstrates a typical inference workflow: starting with global exploration via IF2, followed by gradient-based local refinement, and concluding with a final likelihood evaluation.

```
rw_sd = pp.RWSigma(
    sigmas={
        "mu": 0.0001,
        "kappa": 0.01,
        "theta": 0.00001,
        "xi": 0.001,
        "rho": 0.05,
        "V_0": 0.001,
    },
    init_names=["V_0"],
)
spx_obj.mif(rw_sd=rw_sd, J=1000, M=10, a=0.5, key=subkey)
```

```

eta = {name: 0.001 for name in spx_obj.canonical_param_names}
spx_obj.train(J=1000, M=10, eta=eta, optimizer="Adam")

spx_obj.pfilter(J=1000, reps=10, key=subkey)

print(spx_obj.traces().tail())

```

	theta_idx	iteration	method	logLik	mu	kappa	theta	xi	rho	V_0
17	0	17	train	NaN	NaN	NaN	NaN	NaN	NaN	NaN
18	0	18	train	NaN	NaN	NaN	NaN	NaN	NaN	NaN
19	0	19	train	NaN	NaN	NaN	NaN	NaN	NaN	NaN
20	0	20	train	NaN	NaN	NaN	NaN	NaN	NaN	NaN
21	0	20	pfilter	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Each method call generates a result object (e.g., `PompPFilterResult`, `PompMIFResult`, `PompTrainResult`) which contains execution time, parameter traces, and log-likelihood estimates. These objects are derived from a common base class and provide helper functions like `results()`, `traces()`, and `print_summary()`. This unified architecture ensures that complex multi-stage inference workflows are easily manageable.

4 Discussion

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